Jacob D. Holt, Ph.D. Candidate

https://jakeh15.github.io/

Education

2020 – Present **Dartmouth College**
Ph.D. in Biology

2017 – 2020 **Georgia Institute of Technology (GPA: 3.33/4.0)**
B.S. in Biology with Research Option
Thesis title: *Effect of Hydrodynamic Regime on Snowflake Yeast Evolution*

2016 – 2017 **Georgia Southern University (GPA: 4.0/4.0)**

Publications

1. G. Doing, A. J. Lee, S. L. Neff, *et al.*, "Computationally efficient assembly of pseudomonas aeruginosa gene expression compendia," *mSystems*, vol. 8, no. 1, J. A. Gilbert, Ed., e00341–22, Feb. 23, 2023, ISSN: 2379-5077. DOI: 10.1128/msystems.00341-22. (visited on 03/13/2024).
2. J. D. Holt, D. Schultz, and C. D. Nadell, *Dispersal of a dominant competitor can drive multispecies coexistence in biofilms*, Nov. 28, 2023. DOI: 10.1101/2023.11.28.569042. (visited on 03/13/2024).

Presentations

Talks

2024
- Growth on chitin alters *Vibrio cholerae* biofilm architecture and susceptibility to *Bdellovibrio bacteriovorus* predation
  BALO Symposium, Virtual.
- Dispersal of a dominant competitor can drive multispecies coexistence in biofilms.
  Boston Bacterial Meeting. Boston, MA
- Biofilm ecology of *V. cholerae* on chitin.
  M2P2 Retreat. Lake Morey, VT.

2022
- Mediating social microbes.
  Dartmouth College 3MT. Hanover, NH

Selected Poster Abstracts

2024
- Competitive advantage of parental lysogens in biofilms.
  M2P2 Retreat Lake Morey, Vermont
  J. Peng, **J. Holt**, A. Dalia, and C. Nadell

2023
- Dispersal of a dominant competitor can modulate coexistence in multispecies biofilms.
  GRC Microbial Population Dynamics Proctor, New Hampshire
  J. Holt and C. Nadell

2022
- Dispersal of a dominant competitor can modulate coexistence in multispecies biofilms.
  ASM Biofilms Charlotte, NC.
  J. Holt and C. Nadell
- Growth of *E. coli* and *E. faecalis* on a titanium bead model.
  ASM Biofilms Charlotte, NC.
  M. Bond, **J. Holt**, and C. Nadell

Awards

2024 **MCB Student Representative Fellowship**
Awarded in recognition of service to the MCB graduate program.

2020 **MCB Award**
Award given to competitive applicants to the MCB graduate program.

President's Undergraduate Research Award
Funding awarded by the Georgia Institute of Technology for undergraduate thesis project.

2016 **Presidential Scholar**
Awarded by Georgia Southern University.

2016-2020 **HOPE Scholar**
Tuition scholarship awarded by the University System of Georgia.

Professional Work Experience

2022 **Genetics teaching assistant**
Biological Sciences, Dartmouth College Molecular and Cellular Biology Ph.D. Program

2019 **Undergraduate research assistant**
Yunker Lab, Georgia Institute of Technology

2016 – 2017 **Math and science tutor**
Armstrong TRiO, Georgia Southern University

Committees and Service

Dartmouth College

2024 – 2025 **Graduate Student Mentor**
Científico Latino

Graduate Student Union Steward
GOLD EU

2023 – 2024 **Biology student representative to the graduate committee**
Dartmouth College Molecular and Cellular Biology Ph.D. Program
Responsibilities included; serving on the admissions committee, organizing recruitment weekends, organizing the MCB retreat, and voting on elective courses and faculty appointments to MCB.

2021 – 2023 **Peer mentor**
Dartmouth College Molecular and Cellular Biology Ph.D. Program

2020 – 2022 **Peer research mentor**
Dartmouth College Geisel School of Medicine.

Georgia Institute of Technology

2018-2020 **Undergraduate research ambassador**
Georgia Institute of Technology Vertically Integrated Projects.

Non-Professional

2024-2025 **Executive team member**
Dartmouth Cycling Team.

2019-2020 **Secretary**
Georgia Institute of Technology Cycling Club.

Committees and Service (continued)

Peer mentor
Mission Source Devo Junior Cycling Team.

2018-2020 **Student representative**
USA Cycling South Eastern Collegiate Cycling Conference.

2018-2019 **Vice president**
Georgia Institute of Technology Cycling Club.

2017-2018 **New member coordinator**
Georgia Institute of Technology Cycling Club.

Society Memberships

2021 – Present **American Society of Microbiologists**